Sanskriti

Kumarisanskriti889@gmail.com | LinkedIn | GitHub | Portfolio

Education

Scaler School of Technology **Aug 2024 – Present**
(In collaboration with BITS Pilani) ***CGPA: 9.05*** *Bengaluru, India*

Experience

Code-X-Novas **Sep 2025 – Nov 2025**
Software Developer Intern *Bengaluru, India*

- Developed responsive React.js components for a company website serving 100+ users, translating UI/UX requirements into functional interfaces.
- Integrated frontend features with REST APIs and contributed to authentication configuration for secure, end-to-end application workflows.

Projects

AgentEval – AI Agent Evaluation Platform | Live

Technologies: Python, FastAPI, PostgreSQL, Redis Streams, Docker, AsyncIO

- Designed an AI agent evaluation platform for controlled execution, monitoring, and reproducible benchmarking of agent workflows.
- Implemented asynchronous APIs and workers using FastAPI, PostgreSQL, and Redis Streams, with sandboxed tools and configurable execution limits.
- Built evaluation workflows with task validation, failure classification, execution traces, isolated execution, and structured benchmark scoring.

RepoLens – Source Code Analysis Platform | Live

Technologies: Python, FastAPI, PostgreSQL, Tree-sitter, Docker, TypeScript

- Built a static code analysis platform that parses Python repositories and models functions, classes, imports, call relationships, inheritance, and test references.
- Implemented repository ingestion, immutable snapshots, file inventories, content hashes, language metadata, and unsafe-path detection.
- Developed static name resolution and typed dependency graphs with provenance and confidence labels, plus APIs for symbol search, caller/callee lookup, dependency traversal, and import-cycle detection.

KawaiiLM – AI-Powered Document Question Answering | Live

Technologies: Next.js, React, TypeScript, Express.js, Hugging Face, Qdrant, LangChain

- Built a RAG application for document question answering with text extraction, chunking, and BAAI/bge-large-en-v1.5 embeddings stored in Qdrant for semantic search.
- Integrated Qwen2.5-7B-Instruct through Hugging Face Inference API and implemented SSE streaming with page-level document citations.

VisualSearch – Multimodal Semantic Image Retrieval | Live

Technologies: Python, PyTorch, CLIP, DINOv2, FAISS, Qdrant, FastAPI, React, ONNX Runtime

- Built a multimodal image retrieval system supporting natural-language, image-example, and hybrid queries with metadata filtering using CLIP and DINOv2.
- Implemented FAISS vector search, score calibration, late fusion, and BLIP reranking; achieved 64.7 R@1 with CLIP ViT-L/14 and 83.2 R@1 with BLIP on Flickr30k-1K.

VisionTrack – Object Detection, Tracking & Video Analytics

Technologies: Python, YOLO11, ByteTrack, OpenCV, ONNX Runtime, TrackEval, Docker

- Built a video analytics pipeline using YOLO11, custom ByteTrack, and OpenCV for multi-object tracking, object counting, zone monitoring, and dwell-time analysis across videos and live streams.
- Evaluated tracking performance on MOT17 using TrackEval and pycocotools; optimized inference with ONNX Runtime, quantization, and latency profiling for CPU/edge deployment.

Technical Skills

Programming Languages: C++, Java, Python, TypeScript, JavaScript, SQL
Backend Development: FastAPI, Node.js, Express.js, Spring Boot, REST APIs, Asynchronous Programming, JWT Authentication, BullMQ
Frontend Development: React.js, Next.js, Tailwind CSS, HTML5, CSS3, Jetpack Compose
Databases and Storage: PostgreSQL, MongoDB, Redis, SQLite, Qdrant, Prisma, Firebase, Firestore
Systems and Architecture: Data Structures and Algorithms, OOP, Operating Systems, Database Internals, Query Optimization, Distributed Systems, Concurrency, Caching
AI and Machine Learning: RAG, PyTorch, Hugging Face Transformers, LangChain, Embeddings, NLP, Computer Vision, YOLOv8, ResNet-50
Developer Tools and Testing: Git, GitHub, Linux, Docker, CMake, pytest, Vitest, Playwright, Catch2, Vercel, Render

Achievements

- **Rank 17:** Apple Developer Academy, University of Naples Federico II, Italy.
- **OpenEnv Meta Hackathon:** Placed among the top 800 teams.
- **Finalist:** Smart India Hackathon (SIH 2025).